Lecture 22

Chern classes and the Dirac monopole

Objectives. *A connection can change while its bundle stays fixed. We extract a cohomology class from its curvature that survives this change. On a sphere its integral equals the transition winding and is therefore an integer. For a charged particle in a monopole field, this integer gives Dirac's quantization condition. The tangent sphere supplies a second example through Gauss–Bonnet.*

By the end of this lecture you will be able to:

- Prove that a line bundle's curvature class is independent of its connection.
- Recover its sphere Chern number from the oriented equatorial transition.
- Derive Dirac quantization from the two smooth monopole potentials.
- Relate the tangent sphere and the degree-n line to their Chern numbers.

22.1 The curvature class of a line bundle

Let $E \to M$ be a complex line bundle with a Hermitian fibre metric and a unitary connection. In a unitary frame e_i, the connection's imaginary potential A_i has curvature $F_i = \mathrm{d}A_i$. Lecture 21's frame law on overlaps is

$$e_j = e_i g_{ij}, \qquad A_j = A_i + g_{ij}^{-1}\mathrm{d}g_{ij}, \qquad g_{ij} \colon U_i \cap U_j \to \mathrm{U}(1).$$

Locally write $g_{ij} = e^{\mathrm{i}\chi}$. Then $\mathrm{d}(g_{ij}^{-1}\mathrm{d}g_{ij}) = \mathrm{d}(\mathrm{i}\mathrm{d}\chi) = 0$. Thus the F_i agree and define a global imaginary two-form F. It is closed because $\mathrm{d}F_i = \mathrm{d}^2 A_i = 0$.

For a smooth map $f \colon N \to M$, the *pullback bundle* $f^*E \to N$ has fibre $E_{f(y)}$ at $y \in N$. Pull back the product charts to $f^{-1}(U_i)$. Their transitions $g_{ij} \circ f$ satisfy the same cocycle, so Lecture 20 makes this a smooth bundle.

A *characteristic class* assigns a cohomology class to a bundle, invariant under bundle isomorphism and compatible with pullback. Here the *real first Chern class* is

$$c_1(E) = \left[\frac{\mathrm{i}}{2\pi}F\right] \in H^2_{\mathrm{dR}}(M;\mathbb{R}).$$

The form inside the brackets is real since F is imaginary. We use de Rham classes throughout this lecture. Torsion classes, those killed by a nonzero integer, are not detected by these real forms. The integer sphere computation below needs no torsion comparison.

Theorem 22.1 (Chern–Weil · *cited; line-bundle first Chern class proved given partitions of unity*)**.** *Every complex line bundle has a Hermitian metric and a unitary connection. The class $c_1(E)$ is independent of both choices and is a characteristic class. More generally, on a unitary rank-r bundle, the forms $\mathrm{tr}(F^k)$ are closed and their classes in $H^{2k}_{\mathrm{dR}}(M;\mathbb{C})$ are independent of the connection, for every integer $k \geq 1$. Here powers use matrix wedge multiplication. This general assertion is cited.*

Proof for the line-bundle class. Lecture 12's cited partitions of unity supply the auxiliary choices. Take a locally finite partition ψ_i subordinate to product charts U_i. For $v, w \in E_x$, the weighted sum of their standard chartwise Hermitian products is

$$\langle v, w \rangle = \sum_i \psi_i(x) \langle v, w \rangle_i.$$

It is positive definite since the nonnegative weights sum to one. Their supports and local finiteness make the sum smooth. Normalize the local frames for this metric. In each unit frame let $\nabla^{(i)}$ be the connection with potential zero. The sum $\nabla s = \sum_i \psi_i \nabla^{(i)} s$ defines a global connection. Each summand extends smoothly by zero. The sum is locally finite. The Leibniz and metric product rules are affine in the connection, so their weighted sums give the same rules because $\sum_i \psi_i = 1$. Thus this connection is unitary.

Now compare connections unitary for possibly different metrics. Use the same complex frames for both, without requiring them to be unitary. The line frame law also holds for $h \in \mathbb{C}^\times$. Locally write $h = e^{b+\mathrm{i}\chi}$ with b and χ real. Then $h^{-1}\mathrm{d}h = \mathrm{d}b + \mathrm{i}\mathrm{d}\chi$ is closed, so curvature is unchanged in any such frame. Write the two potentials as $A_i^{(0)}$ and $A_i^{(1)}$ in these common frames. Their difference $B_i = A_i^{(1)} - A_i^{(0)}$ has no transition derivative term, so $B_j = B_i$. These complex one-forms glue to a global B. Both curvatures are imaginary. Taking real parts in their exact difference gives

$$F^{(1)} - F^{(0)} = \mathrm{d}B, \qquad \frac{\mathrm{i}}{2\pi} F^{(1)} - \frac{\mathrm{i}}{2\pi} F^{(0)} = \mathrm{d}\,\mathrm{Re}\left(\frac{\mathrm{i}}{2\pi} B\right).$$

Their real de Rham classes coincide. For a smooth map $f\colon N \to M$, the pullback bundle has pulled-back local potentials. Its curvature is f^*F because pullback commutes with d. Thus $c_1(f^*E) = f^* c_1(E)$. An arbitrary complex bundle isomorphism transports the metric and connection together. Independence of both choices then proves invariance even if the isomorphism does not preserve a previously chosen target metric. ∎

The general statement is Nakahara, Theorem 11.1 and (11.20)–(11.21). No Riemannian metric or orientation on the base entered this definition. For a closed oriented surface Σ, the bundle's *first Chern number* is

$$C_1(E; \Sigma) = \frac{\mathrm{i}}{2\pi} \int_\Sigma F.$$

Changing the surface orientation reverses this number. Stokes makes it independent of the representative curvature in the same class.

22.2 Integrality on the sphere

Give S^2 its outward orientation, equal to the complex orientation of $\mathbb{C}\mathrm{P}^1$. Let D_N and D_S be the closed northern and southern hemispheres. Their common equator C has increasing azimuth ϕ as ∂D_N. The southern boundary is $-C$.

Theorem 22.2 (The sphere Chern number · *proved*). *Let $E \to S^2$ be a Hermitian line bundle with a unitary connection. Choose smooth unitary frames over neighbourhoods of D_N and D_S. In the frame order $e_S = e_N g_{NS}$,*

$$C_1(E; S^2) = \frac{\mathrm{i}}{2\pi} \int_{S^2} F = \frac{1}{2\pi\mathrm{i}} \oint_C g_{NS}^{-1} \mathrm{d} g_{NS} = \mathrm{wind}(g_{NS}) \in \mathbb{Z}.$$

The figure represents the hemispheres as oriented disks.

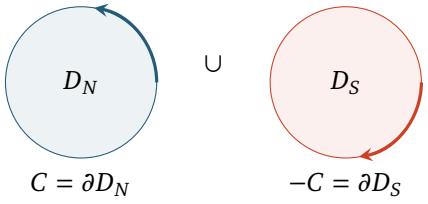


Figure 22.1: The two hemispheres induce opposite orientations on their common equator. Adding their Stokes formulas leaves $\oint_C (A_N - A_S)$. Both disks are drawn as seen from above the north pole.

Proof. Such cap frames exist. Identify the sphere minus the opposite pole with $\mathbb{R}^2$. Choose one unit vector in the fibre over its coordinate origin and transport it along every radial segment to its endpoint. Lecture 21 gives unique unitary transport. Smooth dependence of linear ODE solutions on the endpoint gives a smooth frame. Locally in the endpoint, a finite chain of trivializations covers these compact segments, so this smooth-dependence argument also applies when no initial global frame is given. Each resulting chart contains its entire closed hemisphere.

In these frames, Stokes and the boundary orientations give

$$\frac{\mathrm{i}}{2\pi}\int_{S^2} F = \frac{\mathrm{i}}{2\pi}\left(\int_{D_N} \mathrm{d}A_N + \int_{D_S} \mathrm{d}A_S\right) = \frac{\mathrm{i}}{2\pi}\oint_C (A_N - A_S) = \frac{1}{2\pi\mathrm{i}}\oint_C g_{NS}^{-1}\mathrm{d}g_{NS}.$$

The last equality uses $A_S = A_N + g_{NS}^{-1}\mathrm{d}g_{NS}$ and $-\mathrm{i} = 1/\mathrm{i}$. Lecture 3's circle lifting theorem identifies the last integral with the integer winding. This proves the result for every such bundle, rather than only the monopole potentials. ∎

A different cap frame changes the transition by phases extending over the disks. Each such boundary phase has winding zero, so the integer is unchanged. Lecture 20's cited topological clutching classification applies to the principal $\mathrm{U}(1)$ bundle of unitary frames. It adds that equal integers give continuously isomorphic underlying bundles and that every integer occurs. The tautological line has transition $e^{-\mathrm{i}\phi}$, hence Chern number -1.

22.3 The Dirac monopole

Let $q \neq 0$ be the signed particle charge and Φ_m the magnetic flux. On a sphere enclosing the monopole, the real electromagnetic curvature is

$$F_\mathrm{em} = \frac{\Phi_\mathrm{m}}{4\pi}\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi, \qquad \int_{S^2} F_\mathrm{em} = \Phi_\mathrm{m}.$$

The *Wu–Yang potentials* on the two pole-deleted charts are

$$A_\mathrm{em}^{(N)} = \frac{\Phi_\mathrm{m}}{4\pi}(1-\cos\theta)\mathrm{d}\phi, \quad A_\mathrm{em}^{(S)} = -\frac{\Phi_\mathrm{m}}{4\pi}(1+\cos\theta)\mathrm{d}\phi,$$
$$A_N = -\frac{\mathrm{i}n}{2}(1-\cos\theta)\mathrm{d}\phi, \qquad A_S = +\frac{\mathrm{i}n}{2}(1+\cos\theta)\mathrm{d}\phi,$$

where $n = q\Phi_\mathrm{m}/(2\pi\hbar)$ is initially a real parameter. The ledger is $A = -(\mathrm{i}q/\hbar)A_\mathrm{em}$ and $F = -(\mathrm{i}q/\hbar)F_\mathrm{em}$. The imaginary connection potential includes $q/\hbar$ and is distinct from the real potential.

The polar coordinate ϕ fails at the poles, but these potentials are smooth on their claimed charts. For ambient unit-sphere coordinates (x, y, z),

$$(1 - \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1+z}, \qquad (1 + \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1-z}.$$

The first denominator is nonzero off the south pole, the second off the north pole. Both give the same smooth curvature $F = -(\mathrm{i}n/2)\sin\theta\mathrm{d}\theta \wedge \mathrm{d}\phi$.

Corollary 22.3 (Dirac quantization · *proved*)**.** *The Wu–Yang potentials define a unitary line-bundle connection for charge $q \neq 0$ exactly when*

$$q\Phi_{\mathrm{m}} = 2\pi\hbar n, \qquad n \in \mathbb{Z}.$$

Its sphere Chern number is n.

Proof. On the overlap, $A_S - A_N = \mathrm{i}n\mathrm{d}\phi$. The frame transition must therefore solve $g_{NS}^{-1}\mathrm{d}g_{NS} = \mathrm{i}n\mathrm{d}\phi$. It has the form $ce^{\mathrm{i}n\phi}$ for a constant $c \in \mathrm{U}(1)$. It is single-valued around the equator exactly when $e^{2\pi\mathrm{i}n} = 1$, which holds exactly for $n \in \mathbb{Z}$. For every such integer choose $g_{NS} = (z_{\mathbb{C}}/|z_{\mathbb{C}}|)^n$ on the full overlap, where $z_{\mathbb{C}} = \tan(\theta/2)e^{\mathrm{i}\phi}$ is the complex chart coordinate. Lecture 20 reconstructs its smooth bundle. The connection law glues the two potentials. The sphere theorem gives the bundle's Chern number n. ∎

Here the ambient height z and the complex coordinate $z_{\mathbb{C}}$ are distinct. This is a conditional charge-quantization argument, not an assertion that monopoles exist. The potentials follow Nakahara, §10.5.2, with its imaginary-potential and north/south transition choices translated by the displayed ledger and frame order.

The cap integral need not be an integer. For a northern cap of polar radius $\alpha \in [0, \pi]$, the same curvature gives

$$\frac{\mathrm{i}}{2\pi}\int_{D_\alpha} F = \frac{n}{2}(1 - \cos\alpha).$$

The figure plots $n = 1$. Its hemisphere value is $1/2$, although the sphere number is 1.

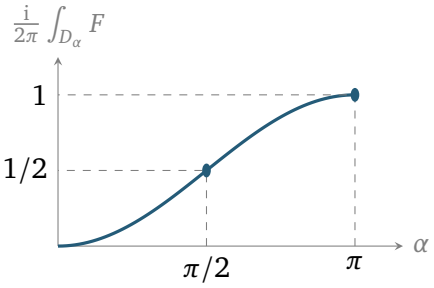


Figure 22.2: For the charge-one curvature, cap flux varies continuously with cap size. Only the full closed-sphere integral is constrained to the integer 1.

22.4 The tangent sphere as a complex line

On an oriented Riemannian surface Σ, let J rotate a positive orthonormal frame (e_1, e_2) by $Je_1 = e_2$ and $Je_2 = -e_1$. Positive orthonormal frame changes commute with this rotation, so it defines multiplication by i on every tangent plane. Its Hermitian fibre product is $\langle u, v\rangle = g(u, v) + \mathrm{i}g(Ju, v)$ for $p \in \Sigma$ and $u, v \in T_p\Sigma$. Lecture 17's Levi-Civita connection

has $\nabla e_1 = -\omega^1{}_2 e_2$ and $\mathrm{d}\omega^1{}_2 = K\mathrm{d}\mathcal{A}$, where $\mathrm{d}\mathcal{A}$ denotes the oriented area form. Its skew connection matrix commutes with J, hence the connection is complex-linear and unitary on this complex line.

Proposition 22.4 (The tangent Chern number · *proved*)**.** *For a closed oriented Riemannian surface Σ, its tangent complex line has*

$$A = -\mathrm{i}\omega^1{}_2, \qquad F = -\mathrm{i}K\mathrm{d}\mathcal{A}, \qquad C_1(T\Sigma;\Sigma) = \frac{1}{2\pi}\int_\Sigma K\mathrm{d}\mathcal{A} = \chi(\Sigma).$$

In particular, $C_1(TS^2;S^2) = 2$.

Proof. The complex frame is e_1, with $e_2 = \mathrm{i}e_1$. Thus $\nabla e_1 = -\mathrm{i}\omega^1{}_2 e_1$, proving the connection formula. Differentiate and use Lecture 17's surface structure equation to obtain the curvature. Multiplication by $\mathrm{i}/(2\pi)$ gives $K\mathrm{d}\mathcal{A}/(2\pi)$. Lecture 18's Gauss–Bonnet theorem evaluates its integral as $\chi(\Sigma)$. For the sphere the Euler characteristic is 2, as required. ∎

This agrees with Lecture 20's tangent-frame winding +2. The inverse component transition winds −2. It is not the transition used in the Chern-number theorem.

22.5 Higher Chern classes

For a unitary rank-r bundle, use the defining $r \times r$ matrix curvature F. The frame law $F' = h^{-1}Fh$ and conjugation invariance of the determinant make the following form global. Even-degree forms commute as scalar entries, so its *total Chern form* is defined by

$$c(F) = \det\left(I + \frac{\mathrm{i}F}{2\pi}\right) = 1 + c_1(F) + c_2(F) + \cdots + c_r(F).$$

The component $c_j(F)$ has degree $2j$ and vanishes when $2j > \dim M$. The corresponding cohomology classes are the *Chern classes*. The invariant-polynomial form of Chern–Weil, Nakahara Theorem 11.1, gives their closedness and connection independence. It applies to each determinant coefficient, not only the traces of powers. For two connections, their difference $B = A^{(1)} - A^{(0)}$ is an endomorphism-valued one-form. With $A_t = A^{(0)} + tB$ and curvature F_t, the cited relative transgression is

$$\mathrm{tr}\big((F^{(1)})^k\big) - \mathrm{tr}\big((F^{(0)})^k\big) = \mathrm{d}\left(k\int_0^1 \mathrm{tr}(B \wedge F_t^{k-1})\mathrm{d}t\right).$$

The trace expression inside is global, although the individual potentials are local. Nakahara's (11.20)–(11.21) supplies this assertion. Expanding the determinant through degree four gives

$$c_1(F) = \frac{\mathrm{i}}{2\pi}\,\mathrm{tr}\,F, \qquad c_2(F) = \frac{\mathrm{tr}(F\wedge F) - \mathrm{tr}\,F \wedge \mathrm{tr}\,F}{8\pi^2}.$$

This uses the elementary determinant coefficient $((\mathrm{tr}\,X)^2 - \mathrm{tr}(X^2))/2$ with $X = \mathrm{i}F/(2\pi)$. For an $\mathrm{SU}(r)$ connection, $\mathrm{tr}\,F = 0$, so $c_2(F) = \mathrm{tr}(F\wedge F)/(8\pi^2)$. Lecture 23 uses this normalization for its second Chern number.

22.6 The degree-n holomorphic line

In a complex chart $z = x + \mathrm{i}y$, a smooth function is holomorphic if $\partial_{\bar z} f = 0$, where $\partial_{\bar z} = \frac{1}{2}(\partial_x + \mathrm{i}\partial_y)$. A *holomorphic line bundle* has local frames with holomorphic transition functions. The $(0,1)$ *part* of a complex one-form is its $\mathrm{d}\bar z$ term. Write $\bar\partial f = (\partial_{\bar z} f)\mathrm{d}\bar z$. Use the projective coordinates z and $w = 1/z$ of Lecture 20. Here z denotes the complex coordinate rather than the ambient height. Define the *degree* of a line over $\mathbb{C}\mathrm{P}^1$ to be its Chern number in the complex orientation. Define the *degree-n line* $O(n)$ for $n \in \mathbb{Z}$ by holomorphic frames with

$$e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z^n.$$

The diagram shows their relation to unitary frames.

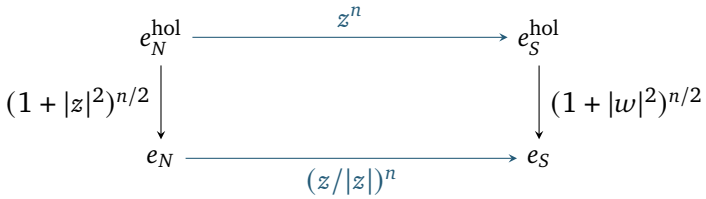


Figure 22.3: Positive real rescalings turn holomorphic frames into unitary frames. Their transition is the phase $(z/|z|)^n$, with winding n.

Example 22.5 (The monopole line as $O(n)$)**.** Identify the Wu–Yang monopole line with $O(n)$, for $n \in \mathbb{Z}$. Equip these frames with squared norms $\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-n}$ and $\|e_S^{\mathrm{hol}}\|^2 = (1+|w|^2)^{-n}$. They agree on the overlap since $|w| = 1/|z|$ and $e_S^{\mathrm{hol}} = e_N^{\mathrm{hol}} z^n$. The unitary frames are

$$e_N = (1+|z|^2)^{n/2} e_N^{\mathrm{hol}}, \qquad e_S = (1+|w|^2)^{n/2} e_S^{\mathrm{hol}}.$$

Their transition is $(z/|z|)^n$. In the northern unit frame the Wu–Yang connection is

$$A_N = -\frac{n}{2}\frac{\bar z \mathrm{d}z - z\mathrm{d}\bar z}{1+|z|^2}.$$

Changing to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-n/2} e_N$ gives

$$A_N^{\mathrm{hol}} = A_N - \frac{n}{2}\mathrm{d}\log(1+|z|^2) = -\frac{n\bar z}{1+|z|^2}\mathrm{d}z.$$

In the southern holomorphic frame the same calculation gives

$$A_S^{\mathrm{hol}} = -\frac{n\bar w}{1+|w|^2}\mathrm{d}w = A_N^{\mathrm{hol}} + n\frac{\mathrm{d}z}{z} \quad \text{on the overlap.}$$

Both expressions have no antiholomorphic differential term and are smooth at their own chart origin. Thus the $(0,1)$ part of the covariant derivative is ordinary $\bar\partial$ in these frames. A local section is holomorphic exactly when its holomorphic-frame coefficient satisfies $\partial_{\bar z} f = 0$. This equips the monopole line with the displayed holomorphic structure. Its Chern number is n by the equatorial winding theorem. For the coefficient ψ in the unit frame the same equation is $(\partial_{\bar z} + nz/[2(1+|z|^2)])\psi = 0$. This local equation will reappear in Lecture 28 with its stated spinor twist.

The cited holomorphic classification on $\mathbb{C}\mathrm{P}^1$ says that every holomorphic line is isomorphic to exactly one $O(n)$, with n its degree. See Gunning, *Function Theory on Compact Riemann Surfaces*, Theorem 2.2, (2.3) and Corollary 2.3, pp. 34–35. This is a holomorphic result in addition to Lecture 20's continuous clutching classification. The tautological frames $(1, z)^T$ and $(w, 1)^T$ have holomorphic transition z^{-1}. The tangent coordinate frames have holomorphic transition $-z^2$ by Lecture 20. Their degrees and the cited classification identify the tautological line with $O(-1)$ and the tangent line with $O(2)$.

Self-check. Show that a trivial Hermitian line bundle over a closed oriented surface has Chern number zero.

Answer. Choose a global unit frame, so the connection has a global one-form A and $F = \mathrm{d}A$. Stokes gives $\int_\Sigma F = \int_{\partial\Sigma} A = 0$ because $\partial\Sigma$ is empty.

Summary. A line bundle's real first Chern class is $[\mathrm{i}F/(2\pi)]$ and is independent of the connection. On the outward sphere its integral is the winding of the frame transition $e_S = e_N g_{NS}$ along the northern boundary. The monopole condition is $q\Phi_\mathrm{m} = 2\pi\hbar n$ with $n \in \mathbb{Z}$. The tangent sphere has Chern number 2, while $O(n)$ has number n. A given complex bundle's class needs no base metric or orientation. The tangent example uses an oriented metric to define its complex structure. Signed surface integrals need the orientation. Lecture 23 uses the second Chern form to study nonabelian fields.

Reading. Nakahara, §10.5.2 and §11.1–11.2. Tu, *Differential Geometry*, §25.8, p. 235. Gunning, Theorem 2.2, (2.3) and Corollary 2.3, pp. 34–35, for the holomorphic classification.